

Q. Write code that when a user enters value in edittext view then display that value when user press button.

• XML

< LinearLayout

< EditText

android:id="@+id/et-name"

^{layout-}

android:width="match-parent"

^{layout-}

android:height="match-parent"

^{or 200dp}

android:hint="Enter your name"

/>

or "Enter Value"

< Button

android:id="@+id/btn-1"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:text="Show"/>

</LinearLayout>

• Java

```
public class MainActivity extends  
    AppCompatActivity
```

```
{  
    EditText et_nameBackend;  
    Button btn_showBackend;
```

@Override

```
protected void onCreate (Bundle  
    savedInstanceState)
```

```
{  
    super.onCreate (savedInstanceState)  
    setContentView (R.layout.activity_main)
```

```
    et_nameBackend = findViewById (R.id.et_nameBackend)  
    btn_showBackend = findViewById (R.id.btn_showBackend)
```

```
    btn_showBackend.setOnClickListener (new  
        View.OnClickListener () {
```

@Override

```
        public void onClick (View v)  
        {
```

```
            Toast.makeText (MainActivity.this,  
                "Welcome " + et_nameBackend.getText ()  
                .toString (), Toast.LENGTH_SHORT).show ();
```

```
        }  
    }
```

for display comparison && for showing square.

```
    btn_showBackend.setOnClickListener (new  
        View.OnClickListener () {
```

@Override

```
        public void onClick (View v)
```

```
        {
```

```
            String s;
```

```
            s = et_nameBackend.getText ().  
                toString ();
```

```
            Integer i = Integer.parseInt (s);
```

```
            if (i > 2)
```

```
                Toast.makeText (MainActivity.this,  
                    "Greater Than 2", Toast.LENGTH_LONG).show ();
```


else

```
Toast.makeText(MainActivity.this,
    "Less than 2", Toast.LENGTH_LONG)
    .show();
}
```

for show square:-

```
btn_showBackend.setOnClickListener(new
    View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Integer i = Integer.parseInt(et_nameBackend
                .getText().toString());

            Integer sv = i * i;
            tvSquareBackend.setText(String.valueOf
                (sv));
        }
    });
```

Q. Show Table using toast.

```
btn_backend.setOnClickListener(new OnClickListener() {
    public void onClick(View view) {
```

```
        String s;
        s = et_backend.getText().toString();
        Integer x = Integer.parseInt(s);
        DisplayTable(x);
    }
```

```
});
}
```

```
public void DisplayTable(int num)
{
```

```
    StringBuilder table = new StringBuilder();
    for (int i = 1; i <= 10; i++)
    {
```

```
        int res = num * i;
        table.append(num).append("*").append(i).
            append(" = ").append(res).append("\n");
    }
```

```
    Toast.makeText(MainActivity.this, table.toString(),
        Toast.LENGTH_LONG).show();
}
```


StringBuilder class:

⇒ String Builder objects are like String objects, they can be modified. Internally these objects are treated like variable-length arrays that contain a sequence of characters. At any point, the length and content of the sequence can be changed through method invocations.

Example:

```
StringBuilder sb = new StringBuilder();  
sb.append("Greetings");
```

toString():

- Used to represent an object as a string.
- it returns the string representation of an object.
- If we print any object, the java compiler internally invokes the toString() method on that object.

getText():

- ⇒ It returns the text from the single-line text field.
- ⇒ It returns only the first line of a multi-line text field.
- ⇒ it used to get text from controls like EditText, Button.

setText():

- ⇒ used to set text on controls
- ⇒ it receives a String.